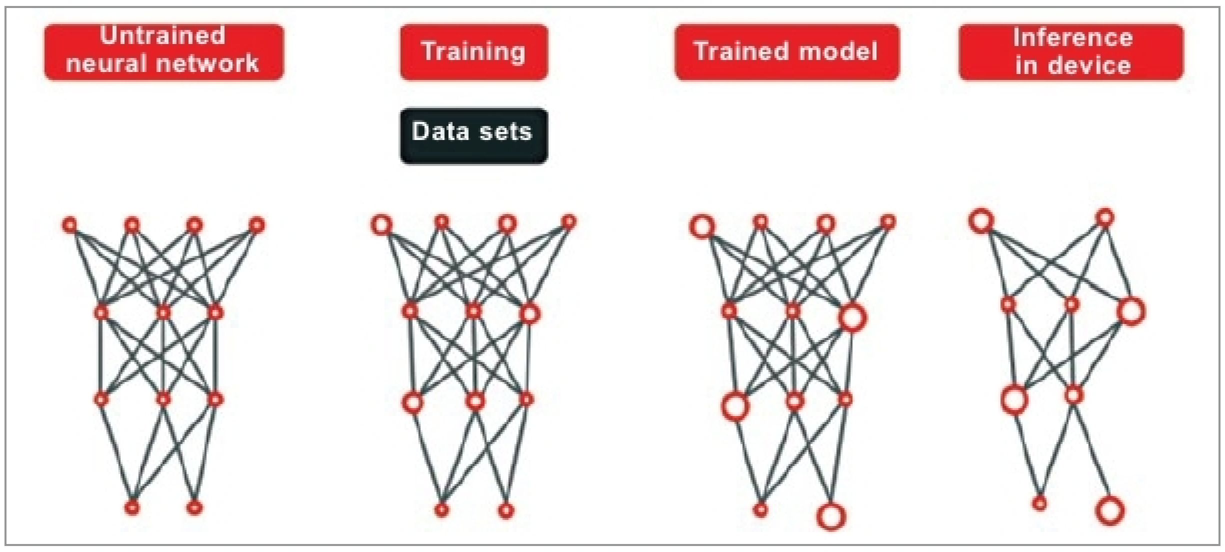

Fig. 2: Schematic illustrating deep learning in data centre/intelligence in the local industrial device (Credit: congatec)

image analysis are always controlled by a doctor and where AI's job is simply to make recommendations, accelerating data evaluation steps in the process.

For this reason, AI systems for use in industrial environments must always ensure that AI decision-making processes are traceable and—as regards machine and occupational safety requirements—100 per cent correct.

Training the AI is therefore much more complex in an industrial environment. There are also hardly any negative examples. This aspect, too, is different from medical technology, where thousands of negative and positive diagnoses can be used to train and teach systems.

In industry, on the other hand, errors must be avoided right from the start. This is why digital twins, that is, digital images of machines and systems, are often used here to simulate negative findings and then, for instance, rule out certain robot movements from the outset. So how do you develop and execute faultless industrial AI solutions for real-time requirements?

Bringing AI to the factory

To get a better understanding, a look at the basics of modern AI systems is helpful. First, it is necessary to distinguish between machine learning (where knowledge is constantly expanded with new clear information) and deep learning (where systems train themselves using large amounts of data and independently interpret new information). The procedure for the latter, that is, deep learning, is almost always identical, even for the most diverse tasks.

Many computing units—usually general-purpose graphics units (GPGPUs)—are combined to form a deep neural network (DNN). This deep learning network must then be trained. In the field of image processing, for example, pictures of various trees can be used to train the system. The amount of image data required is immense. Real-life research projects speak of 130,000 to 700,000 images. Using all this information, neural networks then develop parameters and routines based on case-specific algorithms to reliably identify a tree.

GPGPU: an important technology for AI

In most systems, the main pattern recognition work is done in the GPGPU-based cloud with its immense parallel computing power. However, in the manufacturing industry this is—at least for the time being—ground for exclusion, as processes are generally fast. Here, it must be ensured that intelligence resides at the edge, that is, in the immediate vicinity of or even within the device itself.

This is why industrial devices, machines and systems are mostly equipped with AI systems that use knowledge-based intelligence for real-time applications and pass data for deep learning on to central clouds that cannot be connected in real time yet. This means, it is possible today to keep training a higher-level system with all new data and to keep local devices up to date with regular software updates, so that such systems already classify as self-learning systems.

Only, in this case, learning does not follow a curve but takes place in cyclical steps. This is also a reason why issues such as digital twins or industrial edge servers are so important. If you can provide both, even deep learning can become increasingly real-time capable.

The right embedded processors

It does not matter which setup OEMs choose for their AI. The required, often massive, parallel-processing power for individual real-time capable machines or systems is nevertheless very high, even when using ordinary, purely knowledge-based AI. Latest embedded accelerated processing units (APUs) from companies like AMD support this need, as they offer a powerful GPU in addition to the classic x86 processor, whose general-purpose functions also support parallel AI computing processes, such as those used in data centres.

Using discrete embedded GPUs from the same manufacturer, this can be scaled further, making it possible to adapt open parallel computing performance to the precise requirements of the industrial AI application.

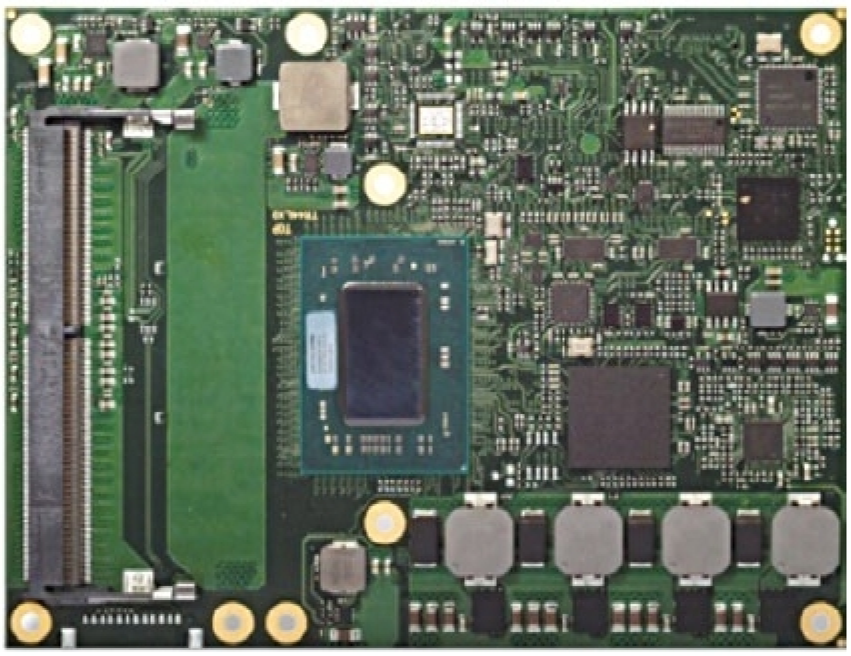

Fig. 3: conga-TR4 with AMD Embedded Ryzen processor (Credit: congatec)